

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA1558

COURSE OUTCOME COVERED

CO4: Analyze big data characteristics and challenges across domains including security and application aspects.
(BL4)

Dave's Taxonomy: Articulation (Level 4)

Question Count: 5

Duration: 60 Minutes

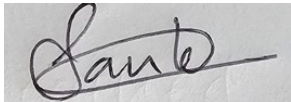
Total Marks: 40

CODE OF CONDUCT

I SANDEEP K / 192521253 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



SANDEEP K / 192521253

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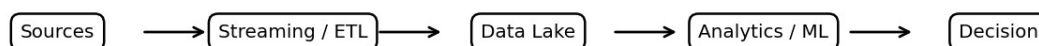
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1. Retail Data Optimization

Retail data comes from physical stores and online channels and must support both immediate inventory visibility and forecasting.

- Stream POS and online events.
- Store historical sales and behavior in a data lake.
- Use batch analytics for demand forecasting.
- Use customer features for personalized recommendations.
- Protect data with encryption and access control.

Industry Big Data Architecture

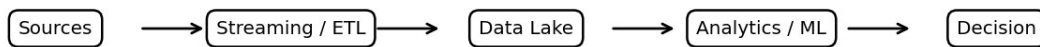


2. Social Media Fraud Detection

Billions of interactions require scalable near-real-time analysis. Streaming features and ML can flag suspicious accounts and content.

- Ingest posts, comments, likes and shares through distributed streaming.
- Extract behavioral and network features in real time.
- Use ML classification or anomaly detection.
- Retain history for model training.
- Monitor false positives, privacy and model quality.

Industry Big Data Architecture



3. Government Data Platform

A national platform needs shared definitions, interoperability and strict governance so departments can use integrated data safely.

- Standardize metadata and common definitions.
- Connect agencies through APIs and interoperable formats.
- Apply role-based access, encryption and audit logs.
- Run quality checks for completeness and duplicates.
- Define ownership, retention, privacy and compliance rules.

Industry Big Data Architecture



4. Banking Fraud Detection

Banking requires low-latency decisions and high accuracy. Combining live transaction signals with historical behavior improves risk scoring.

- Stream transactions through distributed messaging.
- Combine live and historical customer features.
- Use rules plus ML/anomaly detection.
- Generate immediate alerts or step-up verification.
- Maintain high availability, encryption and auditability.

Industry Big Data Architecture



5. Agriculture Smart Farming

Sensor and drone data is distributed across fields and changes continuously. Edge and cloud analytics can turn observations into farming actions.

- Collect soil, weather, crop and drone data.
- Use edge processing for immediate responses.
- Apply time-series and geospatial analytics.
- Predict yield, irrigation and disease risk.
- Deliver actionable dashboards securely.

Industry Big Data Architecture

